

formHTR: Template-Guided OCR Extraction of Handwritten Content from Scanned Forms

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

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Submitted: 11 June 2026

Published: unpublished

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Summary

formHTR is a Python software package for automatic handwritten text recognition (HTR) and extraction from scanned form documents. While the content extraction itself is accomplished using advanced pretrained OCR models, a key step in the process is considering the prior knowledge of the expected contents. That is achieved by the precise, semi-automatic annotation of the regions of interest (ROIs) in the form template, specifying the locations and content types of ROIs. Using this approach in combination with comparing and evaluating outputs from multiple OCR models significantly improves the quality of the extraction.

Statement of need

Large-scale scientific expeditions often collect huge amounts of samples, with a need to record the origin and context of the samples and observed features (metadata). While digital metadata collection methods are becoming more popular, paper forms (so-called *logsheets*) are still the most popular method for their reliability in extreme environments, stability, natural scalability, and ease of use ([Brewer & Guiterman, 2016](#); [van Tamelen, 2004](#)).

It is necessary to build an infrastructure that processes logsheets automatically with a minimal amount of manual interventions and laborious proofreading. Optical character recognition (OCR) methods ([Mori et al., 1992](#)) are used to extract the content from a scanned logsheet, with additional complexity added by the handwritten nature of the content. While training custom models on a particular handwriting style of a person is generally more precise, this approach is impractical for large-scale expeditions with high turnaround of staff and consequently higher amount of distinct handwritings. The use of multiple pretrained general purpose OCR models, allowing consensus or majority decision making, is a more suitable approach.

Just as large-scale expeditions enforce standards on sample collection, they also enforce them on the metadata level. That means standardised logsheets are often developed, and used repeatedly in various sampling scenarios. As a consequence, to digitalise the contents of such logsheets, we can leverage their known structure and expected content types to navigate the OCR methods for more reliable and precise results.

State of the field

The current OCR landscape of pretrained tools can be split between open-source models and cloud-based services. On the open-source side, there are many tools such as Tesseract OCR ([Google & HP, 2021](#)), EasyOCR ([Jaided AI, 2024](#)), OCR4All ([Reul et al., 2019](#)), and PaddleOCR ([Cui et al., 2025](#)) which provide pretrained models or optionally allow to train

39 custom models. Some of the trade-offs are in accuracy, speed, and deployment complexity.
40 Cloud-based pretrained OCR services, including Google Cloud Vision API (Google LLC, 2024),
41 Azure AI Document Intelligence (Microsoft, 2024), and Amazon Textract (Amazon Web
42 Services, 2024), are accessible via APIs and are highly optimised for structured documents and
43 provide higher-level outputs (e.g. key-value pairs).

44 Naturally, the combination of multiple tools, models, or services has emerged. Systems such
45 as OCRmyPDF (Barlow, 2026) or unified interfaces like OcrPy (maxentlabs, 2022) integrate
46 engines like Tesseract, cloud APIs, and downstream processing into a single pipeline, effectively
47 abstracting over multiple OCR backends. The tool Handprint (Caltech Library, 2020) combines
48 multiple cloud services and outputs annotated images or raw results, as well as compares the
49 recognised text against the expected content¹. None of these tools integrates a reusable form
50 template, aligns a scan to that template, assigns recognised fragments to named fields, or
51 outputs a table with fused results from several engines by the voting process. Extending any
52 of them would have meant replacing that output model rather than reusing it, so a dedicated
53 extraction pipeline was a more straightforward design.

54 Software design

55 formHTR is structured into two main steps with an overview in Figure 1. The specification
56 step selects regions of interest (ROIs) from a template and assigns them a meaning. This is
57 achieved by selecting and manipulating the region locations, while the variable names and
58 ROI types are assigned to individual ROIs. The output of the specification step is a config file
59 containing position, name, and type of identified ROIs.

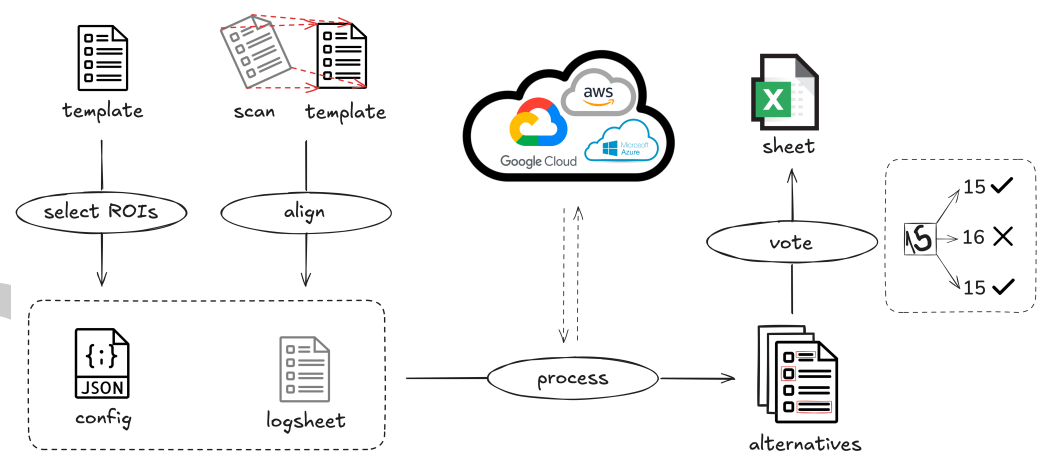


Figure 1: Schematic overview of formHTR annotation and processing workflow.

60 The next step identifies and extracts content from the ROIs. First, the scanned logsheet and
61 its template are aligned to ensure the ROIs actually match the regions in the scanned logsheet
62 (otherwise they would point to potentially empty or generally mismatched regions). While the
63 goal is straightforward, the execution can be problematic, especially when the template has no
64 fiducial markers or the scan quality is low. For this reason, a manual alignment is possible as
65 well.

66 The aligned logsheet is converted to an image and several OCR models are queried to identify
67 and extract the content. For this purpose, three services are used by calling their respective
68 APIs - Google Cloud Vision (Google LLC, 2024), Azure AI Document Intelligence (Microsoft,
69 2024), and Amazon Textract (Amazon Web Services, 2024). Cloud APIs were preferred to

¹The package is not maintained anymore.

open-source engines for handwriting recognition quality and to avoid training or hosting models at expedition scale. Users accept an external dependency, setup and possible usage costs², and the privacy implications of sending scans to third parties; they can enable any subset of the three services, or later add self-hosted backends. All three services output a set of detected words with their location (bounding box) in the image.

After collecting the identified contents for each service, the process of binning assigns each captured text fragment (word) to the appropriate ROI. Several cases need to be considered, such as a word spanning over multiple ROIs, multiple words overlapping with a single ROI, or a sentence split into several words (see examples in Figure 2). To find all overlaps for a ROI from all the services, we use R-tree (Gillies, 2025) to index the regions and capture their overlaps for all services' outputs and the template specification, and consequently use it to effectively query for intersections. Additionally, the logsheet can contain preprinted contents (*residuals*) that do not belong to ROIs, but can be shifted to its boundary box by an imperfect alignment. This is handled by defining the residuals already in the specification step, thus allowing to exclude them from the outputs.

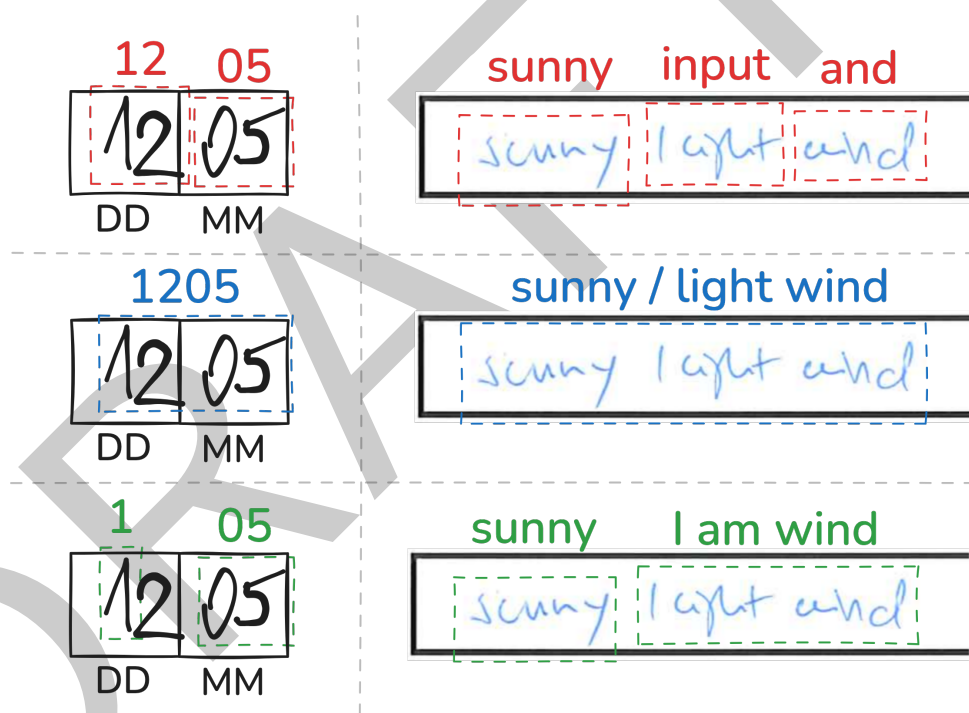


Figure 2: Examples of different scenarios of text position identification relative to a ROI. Each row (distinguished by colour) corresponds to a different service. In the left column, two regions are in close proximity, causing the identified text to overlap with a neighbouring region or to be recognised as a single word. In the right column, the detected words and sentence splitting into words vary across services.

Assuming the words are binned for all services, we can use a voting algorithm to pick the most likely output. With three services, a majority vote can be applied. In cases where there is no consensus, a random choice is made for simplicity. A more elaborate consensus would need comparable confidence scores or labelled data, which we do not assume, thus implementing a richer rule remains a natural future extension. The tool works even with less than three services enabled, albeit with lower output quality due to the inability to perform the voting. On

²All services offer a free tier with limited amount of requests per month. The user needs to provide suitable account credentials. A quick start description how to do this is available at the [wiki pages](#) of the formHTR repository.

91 top of the voting, the weight of votes can be altered by using the known information about the
92 ROIs. In the current version, priority is given to numerical values (if expected). This aspect
93 has a very high potential for extensions in the future versions (e.g. allow only values from a
94 dictionary, only content satisfying a regex, only numbers from an integer range, or content
95 satisfying a length threshold).

96 The output of formHTR is an Excel spreadsheet (an .xlsx file) with two sheets. Metadata
97 sheet has three columns - a variable name coming from the specification, the extracted content
98 as the result of the voting algorithm, and a picture cut out of the scanned (and aligned)
99 logsheet based on the bounding box defined in the specification. This can be used for a quick
100 proofreading of the outputs. Extra sheet contains any miscellaneous content that neither
101 falls into any ROI nor was filtered out as a residual. This can typically alert the user to any
102 comments and notes written on unexpected parts of the logsheet.

103 Research impact statement

104 The need for the formHTR tool arose during the TRaversing European Coastlines (TREC)
105 expedition³. The tool was developed and continuously optimised for the expedition, and
106 consequently used on a collection of approximately ten thousand double-sided logsheets
107 spanning around a hundred different template types, together providing contextual metadata
108 to more than eighty thousand collected samples. Using formHTR, the logsheet templates were
109 annotated and all the scanned logsheets processed. The metadata was extracted and curated,
110 and archived⁴ in the BioSamples database (Courtot et al., 2022). The contextual metadata is
111 an essential foundation for sample analysis and data production, which, at the time of writing,
112 is in the initial phase.

113 Example workflow

114 The following is an example workflow in which we first create and annotate ROIs for the
115 template, and consequently process the scanned logsheet, ultimately creating the output Excel
116 spreadsheet.

```
117 # 1) Create ROI config for a template
118 formhtr select-rois --pdf-file template.pdf --output-file config.json
119
120 # 2) Annotate ROI types and variable names
121 formhtr annotate-rois --pdf-file template.pdf --config-file config.json --
122 output-file config_annotated.json
123
124 # 3) Process a scanned logsheet
125 formhtr process-logsheet \
126 --pdf-logsheet scan.pdf \
127 --pdf-template template.pdf \
128 --config-file config_annotated.json \
129 --output-file output.xlsx \
130 --google google_credentials.json \
131 --amazon amazon_credentials.json \
132 --azure azure_credentials.json
```

³<https://www.embl.org/about/info/trec/>

⁴<https://www.ebi.ac.uk/biosamples/samples?text=Traversing+European+Coastlines+%28TREC%29+expedition>

Author Contributions

MT wrote the manuscript and developed the software. JG contributed to the software. SP, MK, and KL contributed via conceptual guidance and contributed to the manuscript.

AI usage disclosure

Generative AI assistance was used through the Cursor IDE with automatic agent selection, so individual model names and versions were not recorded. It was used to draft tests and documentation, and during JOSS review to help revise this manuscript and author responses. It was not used to design the extraction pipeline or to write the original manuscript. Human authors reviewed, edited, and validated all AI-assisted outputs and made the core design decisions.

Acknowledgements

We thank the late Peer Bork for general supervision of this work. This publication was enabled by the support of EMBL member states to the TREC expedition (within the framework of EMBL's Molecules to Ecosystems Programme 2022-2026) and partially funded by the European Union's Horizon 2020 research and innovation programme (project BIOcean5D with grant agreement No. 101059915). Views and opinions expressed are, however, those of the authors only and do not necessarily reflect those of the European Union. Neither the European Union nor the granting authority can be held responsible for them.

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